



An-Najah National University
Faculty of Engineering and IT

جامعة النجاح الوطنية
كلية الهندسة وتكنولوجيا المعلومات

Distributed Operating Systems

10636456

2nd Semester | 2025-2026

Bazar-com Project | output

Roaa Bzoor	12218040
Yaman Abo Asal	12240055

Once the system is running:

```
[+] up 10/10
✓ Image bazar1com-order          Built
✓ Image bazar1com-order_2        Built
✓ Image bazar1com-frontend        Built
✓ Image bazar1com-catalog        Built
✓ Image bazar1com-catalog_2      Built
✓ Container bazar1com-order_2-1   Recreated
✓ Container bazar1com-order-1     Recreated
✓ Container bazar1com-catalog_2-1 Recreated
✓ Container bazar1com-catalog-1   Recreated
✓ Container bazar1com-frontend-1  Recreated
Attaching to catalog-1, catalog_2-1, frontend-1, order-1, order_2-1
```

```
catalog_2-1 | * Running on all addresses (0.0.0.0)
catalog-1   | * Running on all addresses (0.0.0.0)
catalog-1   | * Running on http://127.0.0.1:5001
catalog_2-1 | * Running on http://172.18.0.5:5001

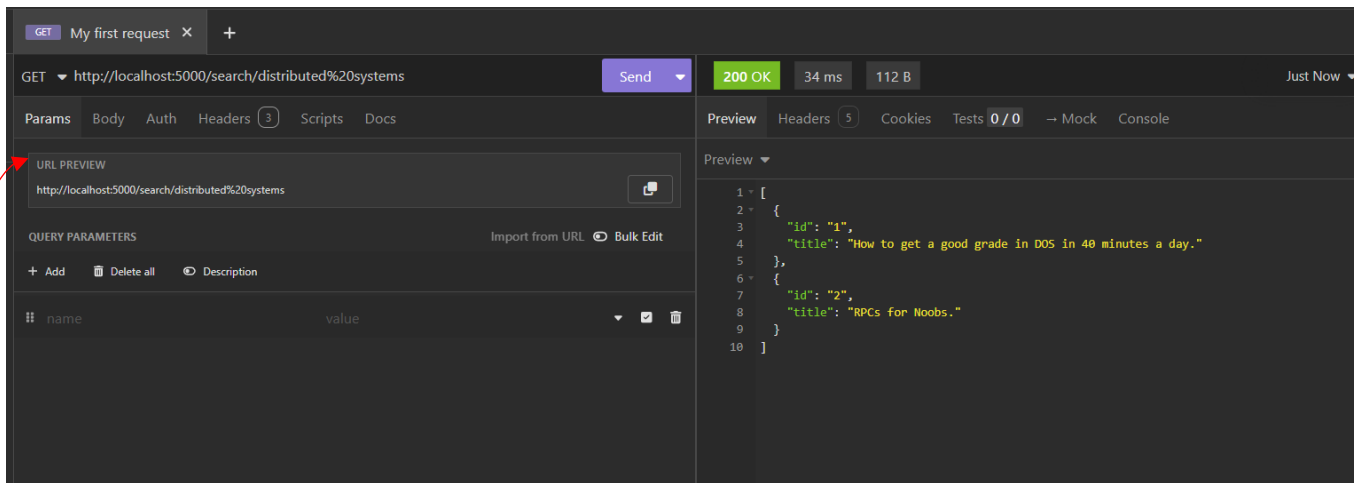
catalog-1   | * Running on http://172.18.0.4:5001
catalog_2-1 | Press CTRL+C to quit
catalog-1   | Press CTRL+C to quit

order-1     | * Serving Flask app 'app'
order-1     | * Debug mode: off
order_2-1   | * Debug mode: off
order-1     | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.

order-1     | * Running on all addresses (0.0.0.0)
order_2-1   | * Running on all addresses (0.0.0.0)

order_2-1   | * Running on http://127.0.0.1:5002
order-1     | * Running on http://172.18.0.3:5002
order_2-1   | * Running on http://172.18.0.2:5002
order_2-1   | Press CTRL+C to quit
frontend-1  | * Serving Flask app 'app'
frontend-1  | * Debug mode: off
frontend-1  | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
frontend-1  | * Running on all addresses (0.0.0.0)
frontend-1  | * Running on http://127.0.0.1:5000
```

Searching for books with topic "distributed systems":



GET My first request × +

GET http://localhost:5000/search/distributed%20systems Send 200 OK 34 ms 112 B Just Now

Params Body Auth Headers (3) Scripts Docs

URL PREVIEW
http://localhost:5000/search/distributed%20systems

QUERY PARAMETERS
+ Add - Delete all Description

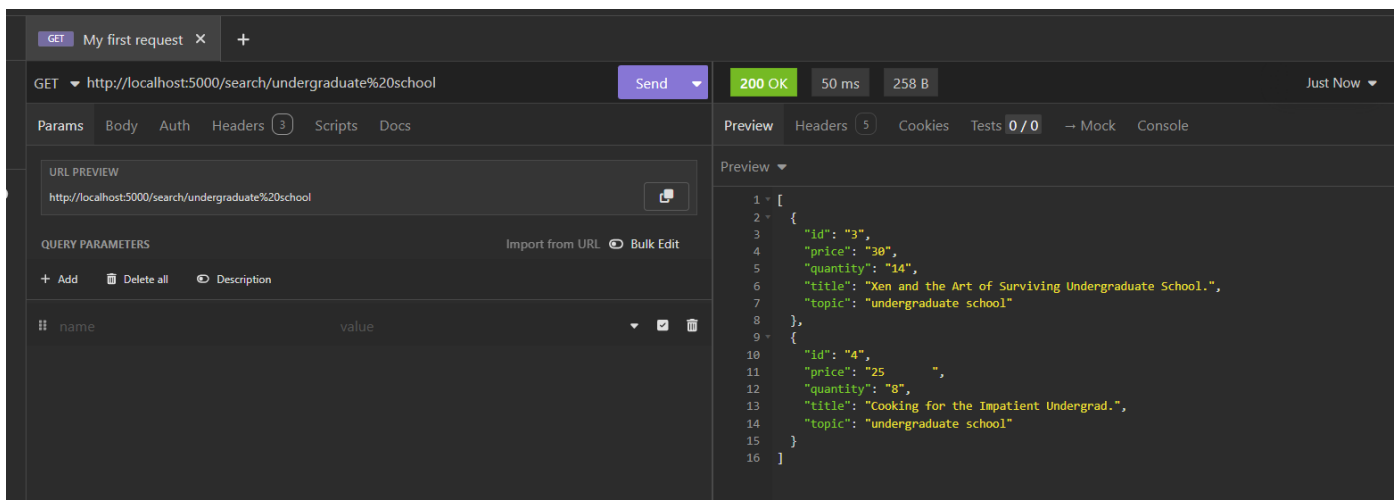
name	value
------	-------

Preview Headers (5) Cookies Tests 0/0 → Mock Console

```
1 - [  
2 - {  
3   "id": "1",  
4   "title": "How to get a good grade in DOS in 40 minutes a day."  
5 },  
6 - {  
7   "id": "2",  
8   "title": "RPCs for Noobs."  
9 }  
10 ]
```

```
frontend-1 | Press CTRL+C to quit  
frontend-1 | Search request for topic: distributed systems  
catalog-1 | 172.18.0.6 - - [11/May/2026 17:28:44] "GET /search/distributed%20systems HTTP/1.1" 200 -  
frontend-1 | SEARCH CACHE MISS  
  
frontend-1 | FORWARDED TO: http://catalog:5001  
frontend-1 | Search successful, found 2 books  
frontend-1 | 172.18.0.1 - - [11/May/2026 17:28:44] "GET /search/distributed%20systems HTTP/1.1" 200 -
```

Searching for books with topic "undergraduate school":



GET My first request × +

GET http://localhost:5000/search/undergraduate%20school Send 200 OK 50 ms 258 B Just Now

Params Body Auth Headers (3) Scripts Docs

URL PREVIEW
http://localhost:5000/search/undergraduate%20school

QUERY PARAMETERS
+ Add - Delete all Description

name	value
------	-------

Preview Headers (5) Cookies Tests 0/0 → Mock Console

```
1 - [  
2 - {  
3   "id": "3",  
4   "price": "30",  
5   "quantity": "14",  
6   "title": "Xen and the Art of Surviving Undergraduate School.",  
7   "topic": "undergraduate school"  
8 },  
9 - {  
10  "id": "4",  
11  "price": "25",  
12  "quantity": "8",  
13  "title": "Cooking for the Impatient Undergrad.",  
14  "topic": "undergraduate school"  
15 }  
16 ]
```

```

frontend-1 | Press CTRL+C to quit
frontend-1 | 172.18.0.1 - - [11/May/2026 17:47:07] "GET / HTTP/1.1" 200 -
frontend-1 | Search request for topic: undergraduate school
catalog-1 | 172.18.0.6 - - [11/May/2026 17:47:17] "GET /search/undergraduate%20school HTTP/1.1" 200 -

frontend-1 | SEARCH CACHE MISS
frontend-1 | FORWARDED TO: http://catalog:5001
frontend-1 | Search successful, found 2 books
frontend-1 | 172.18.0.1 - - [11/May/2026 17:47:17] "GET /search/undergraduate%20school HTTP/1.1" 200 -

```

Get information about book with id=1:

The screenshot shows a web browser's developer tools with a GET request to `http://localhost:5000/info/1`. The response is a 200 OK status with a 50 ms response time and 131 B of data. The response body is a JSON object:

```

{
  "id": "1",
  "price": "20",
  "quantity": "6",
  "title": "How to get a good grade in DOS in 40 minutes a day.",
  "topic": "distributed systems"
}

```

```

catalog-1 | 172.18.0.6 - - [11/May/2026 18:03:26] "GET /info/1 HTTP/1.1" 200 -
frontend-1 | Info request for id: 1
frontend-1 | CACHE MISS
frontend-1 | FORWARDED TO: http://catalog:5001
frontend-1 | Info fetched: How to get a good grade in DOS in 40 minutes a day.
frontend-1 | 172.18.0.1 - - [11/May/2026 18:03:26] "GET /info/1 HTTP/1.1" 200 -

```

Purchase for book with not available id:

The screenshot shows a web browser's developer tools with a POST request to `http://localhost:5000/purchase/5`. The response is a 200 OK status with a 49 ms response time and 27 B of data. The response body is a JSON object:

```

{
  "error": "book not found"
}

```

```

frontend-1 | FORWARDED TO: http://order_2:5002
frontend-1 | INVALIDATING SEARCH CACHE
frontend-1 | 172.18.0.1 - - [11/May/2026 18:05:52] "POST /purchase/5 HTTP/1.1" 200 -

```

Purchase for book with id=2:

POST My first request x +

POST http://localhost:5000/purchase/2 Send 200 OK 71 ms 124 B Just Now

Params Body Auth Headers 3 Scripts Docs

URL PREVIEW
http://localhost:5000/purchase/2

QUERY PARAMETERS
+ Add - Delete all Description Import from URL Bulk Edit

#	name	value
---	------	-------

Preview Headers 5 Cookies Tests 0/0 Mock Console

Preview

```
1 {  
2   "message": "bought book RPCs for Noobs.",  
3   "order": {  
4     "book_id": 2,  
5     "price": "50",  
6     "status": "completed",  
7     "title": "RPCs for Noobs.",  
8   }  
9 }
```

```
catalog-1 | 172.18.0.4 - - [11/May/2026 18:08:59] "GET /info/2 HTTP/1.1" 200 -  
order-1   | 172.18.0.6 - - [11/May/2026 18:08:59] "POST /purchase/2 HTTP/1.1" 200 -  
frontend-1 | FORWARDED TO: http://order:5002
```

Purchase a book out of stock:

POST My first request x +

POST http://localhost:5000/purchase/2 Send 200 OK 32 ms 38 B Just Now

Params Body Auth Headers 3 Scripts Docs

URL PREVIEW
http://localhost:5000/purchase/2

QUERY PARAMETERS
+ Add - Delete all Description Import from URL Bulk Edit

#	name	value
---	------	-------

Preview Headers 5 Cookies Tests 0/0 Mock Console

Preview

```
1 {  
2   "message": "out of stock right now!"  
3 }
```

```
frontend-1 | FORWARDED TO: http://order_2:5002  
order_2-1  | 172.18.0.6 - - [11/May/2026 18:09:29] "POST /purchase/2 HTTP/1.1" 400 -  
catalog-1  | 172.18.0.3 - - [11/May/2026 18:09:29] "GET /info/2 HTTP/1.1" 200 -  
frontend-1 | INVALIDATING SEARCH CACHE  
frontend-1 | 172.18.0.1 - - [11/May/2026 18:09:29] "POST /purchase/2 HTTP/1.1" 200 -
```